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# Addressing Algorithmic Bias in AI-Driven HRM Systems: Implications for Strategic HRM Effectiveness

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## ABSTRACT

AI and machine learning algorithms are revolutionising the modern workplace by transforming HR functions to deliver superior outcomes for both employees and organisations. However, research shows that these algorithms often fail to deliver optimal HR solutions, primarily due to inherent biases. Developing capabilities to overcome algorithmic biases is critical for firms, as these biases present significant challenges to fairness and inclusivity in HR decision-making, ultimately impacting the effectiveness of HR practices. To address this challenge, our study, grounded in the dynamic capability perspective, presents a model to address algorithmic biases in people management and achieve superior strategic HR outcomes. To test our theoretical model, we collected survey data using a two-wave, time-lagged approach from HR professionals and employees working in firms within the Australian financial and insurance industries. The key findings reveal three critical dimensions of HR algorithmic bias management capability: data bias, model bias, and deployment bias management capabilities, which significantly influence AI-enabled high-performance HR practices and, in turn, positively impact strategic HRM effectiveness. Our novel findings on the dimensions of HR bias management capability contribute to advancing the dynamic capability view in HRM research. They also offer a comprehensive bias management framework that allows HR professionals to address the strategic, ethical, and operational challenges emerging from the use of AI-augmented HR practices in the dynamic workplace, helping sustain a competitive advantage.

## 1 | Introduction

Advanced technological capabilities are revolutionising the contemporary world of work, with automation and the augmentation of people management practices taking centre stage (Malik, Budhwar, Mohan, and Nr 2023; Parent-Rocheleau et al. 2024; Pereira et al. 2023). These advancements bring both significant opportunities and challenges for human resource management, particularly in the domain of Algorithmic HRM (AHRM). Also known as AI-driven or AI-augmented HRM, AHRM leverages AI techniques to collect, process, and analyse data, facilitating HR-specific problem-solving and decision-making.

The current rise of AHRM is largely driven by the reported impacts of algorithmic management on organisational outcomes such as cost and time efficiency, productivity, risk management, decision-making, and process consistency and standardisation (Chowdhury et al. 2023; Parent-Rocheleau and Parker 2022), as well as its influence on employee outcomes including motivation, engagement, empowerment, collaboration, and performance (Malik, Budhwar, Mohan, and Nr 2023; Pereira et al. 2023; Rodgers et al. 2023). Hence, the widespread application of algorithms across various HR functions is increasingly witnessed. For instance, Amazon is known for its hiring and firing practices using algorithms (Chen 2023), Uber for dynamic

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**Summary**

- What is currently known?
  - AI and machine learning algorithms have been increasingly streamlining HR functions like hiring, performance management, and termination.
  - Algorithmic biases often fail to deliver fair and inclusive HR decisions.
  - Addressing algorithmic biases is crucial for achieving strategic HR outcomes.
- What does this paper add?
  - Identifies three dimensions of algorithmic bias in HR: data, model and deployment
  - Demonstrates how managing these biases can enhance high-performance HR practices and strategic effectiveness.
  - Provides a framework for developing bias management capabilities in AI-driven HRM.
- The implications for practitioners?
  - HR professionals must address data, model, and deployment biases for fair and inclusive HR decisions.
  - Developing bias management capabilities can improve the implementation of AI-driven HR practices and enhance strategic HR effectiveness.
  - Practitioners can use the HR bias management capability framework to sustain a competitive advantage in dynamic workplaces.

compensation (Cook et al. 2021), and IBM for its AI-driven career development and management (IBM 2021). Both researchers and practitioners have focused on identifying the most effective bundles of complementary HR practices, often referred to as High-Performance HR Practices (HPHRPs). With the extensive use of algorithms, the focus is now shifting toward AI-enabled HPHRPs.

Despite the considerable potential of AHRM to transform workplaces and their inhabitants for the better, these algorithms often fail to meet expectations due to inherent biases. Algorithmic bias is present when AI generates systematically unfair results that favour or disadvantage certain individuals or groups compared to others (Kordzadeh and Ghasemaghaei 2022). For instance, reports have shown that distinctively Black names can result in a 2.1% decrease in the likelihood of employers reaching out compared to distinctively White names (Curry 2023). China's iTutorGroup was recently fined for age discrimination after using AI to exclude people over 60 (Equal Employment Opportunity Commission 2023). Previously, Amazon also faced criticism for gender discrimination in its hiring algorithms (Dastin 2018). Biases can stem from various sources, such as inaccurate or incomplete data, or from algorithmic models compromised by a lack of integrity in their deployment.

Developing capabilities to overcome algorithmic biases is critical for firms, as these biases present significant challenges to fairness and inclusivity in HR decision-making, ultimately affecting the overall effectiveness of HR practices (M. K. Lee 2018; Newman et al. 2020; Vassilopoulou et al. 2024). Additionally, they can raise concerns about trust, managerial agency, and workplace culture, and may also result in costs such as reputational damage and legal repercussions (Newman et al. 2020; Rodgers et al. 2023). In this context, we explore HR Algorithmic Bias Management Capability (HABMC), defined as a firm's ability to identify, mitigate, and prevent biases in HR algorithms and automated decision-making systems, ensuring fair and equitable outcomes for all stakeholders. We aim to address the following research question: what are the dimensions of HR algorithmic bias management capabilities (HABMC) and how do they influence the strategic HRM effectiveness of organisations?

The strategic HRM discourse, along with the resource-based view (RBV), suggests that HR systems or bundles of HR practices help organisations build their human capital, which can create a competitive advantage when these resources are valuable, rare, hard-to-imitate, and non-substitutable (Barney 1991; B. E. Becker and Huselid 2006). Furthermore, as emphasised by dynamic capability theory (Helfat and Peteraf 2003; Teece 2007), resources become strategic when they are leveraged through dynamic capabilities (DCs) to generate value. DCs enable firms to continuously reconfigure, upgrade, and reintegrate their resources and core capabilities, thereby sustaining a competitive advantage. This study focuses on developing organisational capabilities to effectively utilise AHR and manage biases within the dynamically evolving workplace through HABMC. We conducted a two-wave, time-lagged survey to examine the impact of HABMC and its dimensions on AI-enabled HPHRPs and strategic HRM effectiveness from a managerial perspective. Additionally, we explored how employees' turnover intentions moderate the relationship between AI-enabled HPHRPs and strategic HRM effectiveness from an employee perspective.

Our study makes several important contributions to both literature and practice. First, we contribute to the fields of SHRM and DC by introducing a new capability—HABMC. In doing so, we address recent calls for research to integrate the DC perspective with SHRM (Apascaritei and Elvira 2022; Chadwick and Flinchbaugh 2021), develop distinct DCs within HRM (Apascaritei and Elvira 2022; Chatterji and Patro 2014), and build DCs for algorithm-driven decision-making (Gupta et al. 2020; Simón et al. 2024). For example, a recent review (Apascaritei and Elvira 2022) asserted, '[HRM] systems have been extensively analysed in academic research, yet limited attention has been paid to the role of HRM dynamic capabilities (DC) and their impact on resources and practices,' (p. 1) further stating that 'firms can benefit from HRM DC to rapidly build, integrate and reconfigure their workforce and HRM practices' (p. 11).

Drawing on the DC perspective and its application in AI bias management (Akter et al. 2021, 2023), we establish HABMC, arguing that it equips firms to adapt and evolve in dynamic and high-risk workplace environments, such as those shaped by AI-mediated systems. Despite its benefits, the integration of AI in workforce management has sparked significant tensions, particularly regarding discriminatory HR practices. The DC perspective provides a framework to help organisations navigate shifting landscapes and address the conflicting demands they face (Schilke et al. 2018; Teece 2007). HABMC is presented as a higher-order dynamic capability comprising data, model, and deployment bias management capabilities, along with their nine



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subdimensions. This approach is pivotal because research highlights the need to address AI bias at multiple levels (Charlwood and Guenole 2022; Fabbrizzi et al. 2022; Kordzadeh and Ghasemaghaei 2022). By building on bias management capabilities from other disciplines, we empirically establish the HABMC framework for workforce management, making it the first of its kind in this domain.

Second, we provide theoretical explanations and empirical evidence to establish the impact of HABMC on AI-enabled HPHRPs and strategic HRM effectiveness. In doing so, we advance the discourse on the conditions and pathways that enable HRM to play a strategic role (Jiang and Messersmith 2018) and contribute to the body of research suggesting that effective HR systems and bundles of high-performance HR practices can lead to superior organisational outcomes (Boxall and Macky 2009). Our study broadens the strategic scope of HRM by moving beyond its behavioural impacts on employees to examine firm-level capabilities and outcomes—a dimension that previous research has largely overlooked (Apascaritei and Elvira 2022; Chadwick and Flinchbaugh 2021; Jiang et al. 2013), while also exploring DC outcomes beyond firm performance.

Third, a common concern in the HPHRP literature is whether sufficient attention is given to the context, boundary conditions, and complementarities that influence the effectiveness of these HR systems (Gahan et al. 2021). Addressing this, we focus on AI-enabled HPHRPs, an emerging phenomenon in algorithmic workforce management, by examining their antecedents, outcomes, mediating effects, and intervening conditions, such as employee turnover intentions, that may shape their strategic HR outcomes.

From a practical standpoint, this study provides actionable insights for managing algorithmic biases in HR practices, which often hinder organisations from fully realising the operational and strategic benefits of adopting AI-enabled HR systems. The HABMC framework offers a structured approach to mitigate risks such as discrimination, mistrust, and other negative outcomes while promoting more inclusive and equitable workplaces.

## 2 | Theory Development and Hypotheses

### 2.1 | Dynamic Capability Theory and HR Algorithmic Bias Management Capability

As organisations increasingly adopt analytics and AI, they emphasise developing capabilities to utilise these technologies effectively. The dynamic capabilities (DC) approach offers valuable insights for this development. A firm's capabilities, as defined by Helfat and Lieberman (2002), represent its ability to deploy resources for desired outcomes, comprising both tangible and intangible processes unique to each firm and evolving over time. DCs enable organisations to create, expand, and alter their resource base and core capabilities, facilitating rapid adaptation to the changing business environment and sustaining competitive advantage (Teece 2007). DCs include sensing, seizing, and reconfiguration abilities: sensing capabilities allow firms to detect and respond to environmental opportunities and threats; seizing capabilities involve mobilising resources to capture value; and reconfiguration capabilities continuously update and organise assets to stay competitive. Specific capabilities have been identified in the context of AI and analytics, such as predictive analytics capability (Gupta et al. 2020) and analytics empowerment capability (Motamarri et al. 2022). These capabilities illustrate that technology alone is insufficient; dynamic capabilities must integrate technology with organisational culture and human expertise to achieve performance goals.

With AI's growing applications in HR decisions, there is an increasing focus on understanding and managing algorithmic bias. HR algorithmic bias management capability (HABMC) involves identifying, mitigating, and preventing biases in HR algorithms and automated systems to ensure fair outcomes. A comprehensive review of extant literature reveals that AI bias can originate from various sources or manifest at different levels (Charlwood and Guenole 2022; Fabbrizzi et al. 2022; Kordzadeh and Ghasemaghaei 2022). Our conceptual work builds upon and extends the findings of Akter et al. (2023), who explored AI bias management capabilities in the context of industrial marketing. Drawing on insights from multiple disciplines, we establish HABMC in the HR domain, encompassing HR data, model, and deployment bias management capabilities, as discussed below.

### 2.2 | Developing HR Algorithmic Bias Management Capabilities

Algorithmic HR practices heavily rely on existing data for decision-making. However, biased datasets—caused by historical inequalities, underrepresentation, or systemic discrimination—can perpetuate biases, leading to skewed HR recommendations for managers (Hooker 2021). For example, pay inequity can worsen when flawed datasets with historical gender or race biases are used to run compensation algorithms, even if the algorithms are valid. Based on the DC view, we contend that HR data bias management capability (HDABMC) is a dynamic capability that enables organisations to continuously adapt and refine their processes for identifying, mitigating, and correcting biases in HR data. This involves detecting potential biases in HR data and addressing biases in HR data collection, handling, and usage, thereby transforming HR data governance and decision-making processes. HDABMC equips organisations to address potential biases arising from incomplete, inaccurate, or unrepresentative datasets, thereby enhancing an organisation's dynamic capability to systematically address key dataset attributes—completeness, accuracy, and format—ensuring that data used in algorithmic HR systems is fair, inclusive, and representative (Akter et al. 2023). For example, organisations with robust HDABMC can address exclusion bias by ensuring datasets represent diverse demographics, prevent time interval bias by curating temporally balanced data, and mitigate sample selection bias by correcting skewed datasets. Completeness ensures that datasets thoroughly capture relevant attributes of the task or target population. Inadequate completeness can lead to exclusion bias, where underrepresented groups are systematically excluded, exacerbating inequities (Fabbrizzi et al. 2022). Accuracy pertains to the precision of data in reflecting real-world phenomena. Failure to



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ensure accuracy often results in sample selection bias, where skewed data disproportionately influences model outcomes (Batini and Scannapieca 2006). A notable example is Amazon's AI hiring tool, which exhibited sample selection bias by favouring male candidates. This bias emerged because the training data predominantly represented male employees, reflecting historical gender imbalances in the workforce (Chen 2023). Similarly, poor formatting can lead to time interval bias, where datasets collected over limited or unrepresentative timeframes impair decision-making (Quaresmini and Primiero 2023). Hence, by ensuring the completeness, accuracy, and proper formatting of HR datasets, organisations can develop AHR systems that generate fair and ethical outcomes for all employees, regardless of demographic or social differences. Therefore, we argue that HDABMC is a critical component of HR algorithmic bias management capability and hypothesise:

**H1.** *HR data bias management capability is positively associated with HR algorithmic bias management capability.*

HR model bias management capability (HMOBMC) refers to an organisation's ability to address biases in HR models through methodological and procedural guidelines (Akter et al. 2023). Guided by the DC view, we assert that organisations with well-designed HMOBMC—focusing on detecting and mitigating biases in HR models and frameworks—are better equipped to manage algorithmic biases in HR-related decisions effectively. For example, an algorithm-driven predictive model used to forecast employee turnover may inadvertently produce biased suggestions, disproportionately flagging certain groups as 'at risk' for turnover, highlighting the critical role of robust model bias management. Research indicates that HR model bias occurs due to inaccurate configurations or inappropriate methodological selections, which lead to erroneous predictions and skewed outcomes (Kordzadeh and Ghasemaghaei 2022). Key dimensions of HMOBMC include model reliability, model flexibility, and model ambidexterity (Akter et al. 2023). Model reliability relates to the degree of dependability of an AI-HR model, especially in terms of its technical robustness. An outdated HR training recommendation engine might be ineffective if its statistical parameters and rules do not successfully identify relevant training programs and their determinants. This suggests that to avoid model biases, the model should perform consistently and accurately over an extended period, guaranteeing reliability in its forecasts. In this vein, Whang et al. (2021) stress the importance of robust AI-HR models against noisy or poisoned data to enhance fairness.

Model flexibility pertains to the extent to which an AI-HR model is adaptable to diverse requirements and evolving conditions. An effective HR algorithm used for hiring, for instance, should exhibit consistent and reliable performance across various demographic, geographic, and social groups of candidates. Hence, flexibility in model bias promotes adaptability and responsiveness to diverse data sources and contexts (Blow et al. 2024).

Another important aspect of model bias management capability is model ambidexterity, the level at which an HR model can exploit current opportunities while delving into new opportunities in a constantly changing and dynamic environment. For instance, IBM's Career Navigator Tool is an AI-driven solution that provides employees with real-time, personalised career path suggestions and development recommendations (IBM 2021). IBM appears to apply model ambidexterity by enabling the tool to exploit existing knowledge about career paths while exploring new developmental opportunities for employees. Such algorithm-driven solutions assess employees' long-term value and enhance the value proposition through personalised recommendations and services, while avoiding biased decisions that prioritise only short-term outcomes. Against this backdrop, we argue that firms can develop HABMC by improving their HR model bias through methodological and procedural guidelines concentrating on model reliability, flexibility, and ambidexterity. Therefore, we hypothesise that:

**H2.** *HR model bias management capability is positively associated with HR algorithmic bias management capability.*

HR deployment bias management capability (HDPBMC) refers to a firm's dynamic capability to address and mitigate biases in implementing algorithmic HR solutions. HDPBMC comprises three core elements: competence, benevolence, and integrity, ensuring organisations can deploy AI-powered HR solutions responsibly and ethically. By developing HDPBMC, HR managers can adopt reflexive practices to enhance the firm's competence, benevolence, and integrity, ensuring fair and ethical implementation of AHR models through proactive bias management (Leavitt et al. 2024). The competence dimension refers to the extent to which a firm believes it has the necessary skills and abilities (Mayer et al. 1995) to effectively provide AHR services in a dynamic environment. For instance, Amazon's initiative to reduce repetitive motion injuries among warehouse workers exemplifies this. By leveraging algorithms to rotate employees across tasks that engage different muscle groups, Amazon demonstrates competence in dynamically deploying AI to address workplace safety concerns (Ongweso Jr 2021). Benevolence, another critical dimension, emphasises a firm's commitment to prioritising employees' well-being when deploying AHR solutions (Li and Bitterly 2024). Microsoft exemplifies this through its Microsoft Viva Insights, which utilises AI to help employees enhance personal productivity and well-being (Boyd 2024). Such initiatives harness algorithms to improve employee experiences while mitigating the unpredictable and undesirable effects of AI systems. Integrity, the third pillar of HDPBMC, ensures that organisations adhere to ethical principles such as fairness, justice, and transparency when deploying AI solutions (Colquitt et al. 2007; Nichols 2019). For example, LinkedIn's Talent Insights tool supports recruitment by providing market data while anonymising and aggregating sensitive information like gender and race (Allal-Chérif et al. 2021). This ensures hiring decisions focus on skills and qualifications, upholding fairness and justice in AHR deployment. In the AHR context, firms that mitigate deployment biases through competence, benevolence, and integrity typically have a greater dynamic capacity to manage algorithmic biases in their HR systems. Thus, we hypothesise:

**H3.** *HR deployment bias management capability is positively associated with HR algorithmic bias management capability.*



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### 2.3 | Algorithmic Bias Management Capability in HRM, AI-Enabled High-Performance HR Practices and Strategic HRM Effectiveness

HPHRPs refer to the 'complementary bundles' or unique combination of practices (Boxall and Macky 2009; Takeuchi et al. 2007) that influence patterns of interactions between managers and employees, reshaping how employees work and ultimately leading to gains in performance and/or productivity (Kellner et al. 2023, 172). These combinations of practices collectively achieve performance outcomes that individual practices alone cannot and must be aligned with the operating environment, organisation's philosophy, objectives, and other functions (Van De Voorde and Beijer 2015). While consensus on these complementary practices varies, employee selection, development and career opportunities, performance evaluation, and rewards are well-documented in the literature (Takeuchi et al. 2007; Van De Voorde and Beijer 2015).

In an environment where algorithms increasingly drive most HR practices, we focus on AI-enabled HPHRPs. For instance, algorithmic performance evaluations 'appraise, rate or rank workers' performance or productivity, usually in real-time, typically through the calculation of several metrics or quantified indicators' (Parent-Rocheleau et al. 2024, 27) and algorithmic compensation involves utilising algorithms to calculate employees' salaries based on factors such as task completion, performance, and productivity-related data.

In this context, we assert the criticality of HABMC, including data, model, and deployment capabilities, in attaining the desired outcomes of AI-enabled HPHRPs. For instance, Amazon's lack of data bias management capabilities in their hiring tool resulted in gender bias, potentially missing out on valuable talent. Moreover, algorithmic wage discrimination arising from variable pay raises several questions (Dubal 2023). For example, recent research indicates a 7% gender earnings gap among Uber drivers in the US (Cook et al. 2021). Traditional workplaces deploying AI-enabled HR practices must be aware of replicating such discriminations and ensure the building of HABMCs to reap the desired benefits. We argue that HABMC will positively contribute to fair, efficient and sustainable AI-enabled HPHRPs.

**H4.** *HR algorithmic bias management capability is positively related to AI-enabled high-performance HR practices.*

HRM effectiveness refers to the execution and delivery of superior technical and strategic HRM activities to achieve desired organisational outcomes. Huselid et al. (1997) introduced the distinction between technical and strategic effectiveness in HRM. Traditional HRM practices, which are uniform across firms and labelled as 'technical', include areas such as benefits and compensation, recruitment, training, and retirement strategies. In contrast, innovative HRM activities and policies aligned with overall business strategy are identified as 'strategic' and are reflected through indicators such as employee participation and empowerment, teamwork, workforce flexibility, succession planning, and workforce productivity and quality.

Building on previous research demonstrating that HPHRPs complement HR effectiveness (e.g., Richard and Johnson 2004), we argue that AI-enabled HPHRPs, as a technical system, significantly contribute to a firm's strategic HRM effectiveness. This theorisation is supported by e-HRM research, which has established the acceptance, quality, content, and structure of e-HRM practices influence strategic HR effectiveness (Marler and Fisher 2013; Ruel et al. 2007). In the context of AI-enabled HR practices, for instance, IBM's algorithms deliver personalised training for their employees (IBM 2021), and Amazon employs sophisticated monitoring and surveillance systems and algorithms to predict potential injuries among warehouse employees (McIntyre and Bradbury 2022). These practices have implications for employee empowerment, productivity, and quality of work. Given the merits of AI systems in identifying complex patterns in data, deriving insights effectively, and providing personalised and real-time recommendations, AI-enabled HPHRPs can elevate strategic HRM effectiveness. Therefore, we hypothesise:

**H5.** *AI-enabled high-performance HR practices are positively associated with strategic HR effectiveness.*

AI-enabled HPHRPs have the potential to enhance strategic HR effectiveness by improving employee motivation and engagement, organisational alignment, and operational efficiency (Malik, Budhwar, and Kazmi 2023). However, their ability to drive strategic outcomes depends heavily on a motivated and engaged workforce. Research indicates that strategic HRM effectiveness is likely influenced by employee attitudes and behaviours (Kehoe and Wright 2013), particularly employee turnover intention. Employee turnover intention is an individual's conscious and deliberate decision to leave the organisation, often associated with reduced motivation, lower morale, job satisfaction, organisational commitment, and decreased job performance (Tett and Meyer 1993).

When employees perceive HR algorithms as biased—favouring certain behaviours or metrics not equally achievable by all—they may feel unfairly penalised, leading to dissatisfaction and increased turnover (Chowdhury et al. 2023). High turnover intention can cause employees to disengage from organisational objectives, fail to fully utilise advanced HR systems, or even resist AI-enabled initiatives, viewing them as irrelevant to their future career growth. This disengagement can dilute the intended impact of HPHRPs on strategic outcomes, such as fostering a high-performance culture or driving innovation. Consequently, organisations are unlikely to fully realise the benefits of AI tools in enhancing HR effectiveness. For instance, Google's 2019 internal study identified voluntary turnover among key employees as a significant barrier to maintaining productivity and driving innovation. Employees contemplating departure often disengage from critical initiatives, such as AI-driven performance management tools, which require active participation to succeed. Such employees may perceive AI-driven initiatives differently, potentially undermining strategic HR effectiveness. Therefore, understanding this moderating effect offers valuable insights into employee behaviour and organisational outcomes, shedding light on the complex relationship between AI-enabled HPHRPs and strategic HRM effectiveness. Thus, we hypothesise:



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**H6.** *Employee turnover intention will negatively moderate the relationship between AI-enabled high-performance HR practices and strategic HRM effectiveness, such that the positive association between high-performance HR practices and strategic HRM effectiveness will be weak when employee turnover intention is high.*

We also assert that HABMC has both a direct and indirect effect on strategic HR effectiveness. In terms of direct effect, HABMC helps organisations to identify, mitigate, and prevent biases. This fosters perceptions among employees that they are treated fairly and equitably, ultimately contributing to strategic HRM effectiveness for firms, reflected through increased employee empowerment, productivity and other outcomes. For instance, the fair and ethical use of analytics and algorithms have proven to enhance employee empowerment and participation (Motamarri et al. 2022). In this study, we emphasise the indirect effect of HABMC through the mediating role of AI-enabled HPHRPs in enhancing strategic HRM effectiveness. As hypothesised in H4, bias management capabilities are critical for driving AI-enabled HR practices, given their focus on mitigating biases in data, models, and deployment. Therefore, we argue that firms should focus on developing AI-enabled HR practices strengthened through robust bias management capabilities to achieve greater strategic effectiveness in their HR operations. As discussed in H5, AI-enabled HPHRPs can enhance technical HRM effectiveness, ultimately leading to strategic HRM effectiveness due to their distinct yet complementary nature (Richard and Johnson 2004). Hence, we hypothesise that:

**H7.** *AI-enabled high-performance HR practices will mediate the relationship between HR algorithmic bias management capability and strategic HR effectiveness.*

The proposed relationships and hypotheses are illustrated in Figure 1 below.

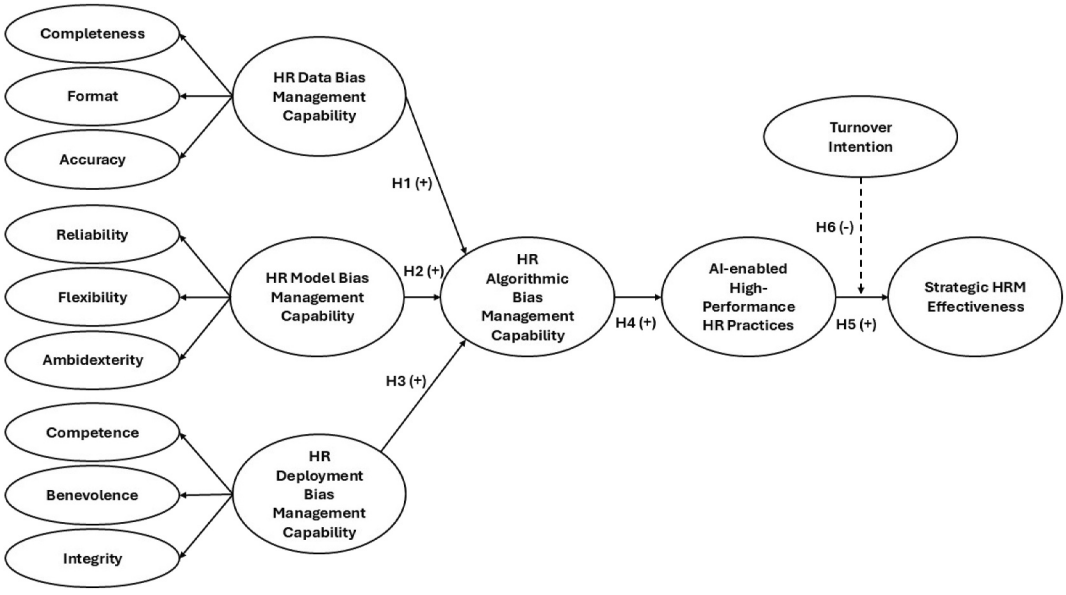



**FIGURE 1** | Research model.

## 3 | Methods

### 3.1 | Research Setting

Our study setting is the Australian financial and insurance industry, a significant contributor to the Australian economy. It actively leverages AI and machine learning algorithms to serve its over 25 million customers and over 544,500 employees, with a share of 48% females (Australian Government 2024). With active partnerships with the leading US tech giants, this sector often leads in adopting AI-empowered tools to optimise people management, fraud detection and prevention, risk management and compliance, and to offer competitive products. For example, the largest bank in Australia, the Commonwealth Bank of Australia, has partnered with Microsoft to foster innovation in AI and cybersecurity within the Australian financial sector (Commbank 2024). Insurance Australia Group (IA) leads in integrating ethical AI into streamlining the HRM processes (NEXT Frontier 2024).

### 3.2 | Measures of Constructs and Pilot Study

We adapted all the measurement scales (see Table 1) previously validated in different contexts and slightly modified them to fit into our study context of leveraging algorithms for people management: data bias management capability (Nelson et al. 2005), model bias management capability (Wixom and Todd 2005), deployment bias management capability (Akter et al. 2023), AI-enabled HPHRP (Takeuchi et al. 2007), strategic HRM effectiveness (Huselid et al. 1997), and employee turnover (Kuvaas and Dysvik 2010). All the constructs were measured using a 7-point Likert Scale. Before the main study, the pretesting phase involved 42 experts, such as HR professionals, data scientists, academics, IT professionals, machine learning experts



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**TABLE 1** | Operationalisation of HR bias management capabilities.

| Constructs | Sub-constructs | Definitions | Item labels | Items |
|---|---|---|---|---|
| HR data bias management capability (HDABMC) | Completeness (COMP) | Refers to the extent to which the dataset provides all the necessary information in a dynamic HR environment | COMP1 | The dataset for HR algorithms provides a complete set of information. |
| | | | COMP2 | The dataset for HR algorithms produces comprehensive information. |
| | | | COMP3 | The dataset for HR algorithms provides all the information needed. |
| | Format (FORM) | Refers to the perception of how well the data is laid out in a dynamic HR environment | FORM1 | The dataset for HR algorithms is well formatted. |
| | | | FORM2 | The dataset for HR algorithms is well laid out. |
| | | | FORM3 | The dataset for HR algorithms is clearly presented on the screen. |
| | Accuracy (ACCU) | Refers to the perceived exactness of the dataset in a dynamic HR environment | ACCU1 | The dataset for HR algorithms produces correct information. |
| | | | ACCU2 | The dataset for HR algorithms contains few errors in the information. |
| | | | ACCU3 | The dataset for HR algorithms provides accurate information. |
| HR model bias management capability (HMOBMC) | Model reliability (RELI) | Refers to the degree to which the model is dependable in a dynamic HR environment | RELI1 | The algorithmic model operates reliably for HR analytics. |
| | | | RELI2 | The algorithmic model performs reliably for HR analytics. |
| | | | RELI3 | The operation of the algorithmic model is dependable for HR analytics. |
| | Model flexibility (FLEX) | Refers to the ability of any AHR model to adapt to a range of user needs and fluctuating conditions in a dynamic HR environment. | FLEX1 | The algorithmic model can be adapted to meet a variety of HR analytics needs. |
| | | | FLEX2 | The algorithmic model can flexibly adjust to new demands or conditions during HR analytics. |
| | | | FLEX3 | The algorithmic model is flexible in addressing needs as they arise during HR analytics. |
| | Model ambidexterity (AMBI) | Refers to the ability to exploit the current opportunities while exploring new ones in a dynamic HR environment. | AMBI1 | The algorithmic model can explore synergies with our existing HR services. |
| | | | AMBI2 | The algorithmic model can specify new strategic possibilities. |
| | | | AMBI3 | The algorithmic model can identify the association between our existing HR services and future ones. |
| HR deployment bias management capability (HDPBMC) | Competence (COMPE) | Refers to the extent to which a firm is believed to have the necessary knowledge and skills to provide | COMPE1 | Our company is competent in providing algorithmic HR services. |
| | | | COMPE2 | Our company performs its HR role very well. |

(Continues)



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**TABLE 1** | (Continued)

| Constructs | Sub-constructs | Definitions | Item labels | Items |
|---|---|---|---|---|
| | | bias-free AHR services in a dynamic environment. | COMPE3 | Our company understands the needs of its employees. |
| | Benevolence (BENE) | Refers to the extent to which a firm is believed to serve the employees with good intentions in a dynamic HR environment. | BENE1 | Our company's algorithmic intentions are benevolent. |
| | | | BENE2 | Our company has good intentions towards its employees. |
| | | | BENE3 | Our company's algorithmic services are well-meaning. |
| | Integrity (INTE) | Refers to the extent to which a firm is believed to commit to moral and ethical principles in a dynamic HR environment. | INTE1 | Promises made by our company are reliable. |
| | | | INTE2 | Our company will keep its commitment. |
| | | | INTE3 | Algorithmic HR services given by our company reflect its best judgement. |

and employees, to validate the manager and employee survey versions. A pilot study with a sample from the target population followed this.

### 3.3 | Main Study

We collected two sources of cross-sectional, time-lagged survey data (with a 6-week interval) from HR professionals (hereafter referred to as managers) who have been leveraging AI algorithms in HR decision-making for at least 3 years, as well as employees in the Australian Finance and Insurance industry. This approach addresses concerns related to temporal dynamics and common method bias, whilst also enabling causal inferences about the relationships between HABMC, AI-enabled HR practices, strategic HRM effectiveness (from a managerial perspective), and turnover intention (from an employee perspective) (Piening et al. 2013; Maynard et al. 2014). This approach aligns with evidence suggesting that HR interventions take time to influence employee attitudes and behaviours, which in turn shape organisational outcomes (Kehoe and Wright 2013). While managers might report positive organisational and employee outcomes from deploying algorithm-enabled HR systems, employees may raise concerns about the fairness of these systems, which could lead to a decline in trust and, in turn, higher turnover. By drawing insights from HR managers and employees, organisations can align AI-driven HR practices with employee expectations, ensuring they are strategically effective, operationally feasible, and positively perceived by employees. Our study delivers robust, actionable insights that enhance theoretical understanding and practical applications through time-lagged methods and multi-level perspectives.

We appointed a reputed Australian online panel data firm to approach the respondents. In round one, out of 679 managers contacted, 387 completed our survey. However, after excluding spurious responses (e.g., straight-lining, missing values, quick response times, and abnormal patterns) (Lee et al. 2024), we considered 274 complete responses to test our theoretical model (see Figure 1). Similarly, in round two, using a simple random technique, we approached 791 employees working in the same industry who experienced AI-driven HR practices such as recruitment and selection, performance management, and promotion decisions. Initially, we collected 419 responses, and after conducting quality checks, 274 were considered for analysis. The appendices present the demographic profiles of both samples: Supporting Information S1: Appendix1 covers managers, and Supporting Information S1: Appendix 2 covers employees, confirming their diversity in terms of gender, age, and experience.

We employed Partial Least Squares based Structural Equation Modelling (PLS-SEM) instead of Covariance-based Structural Equation Modelling (CB-SEM) to better achieve our study's predictive goal (i.e., explaining strategic HRM effectiveness). While CB-SEM is primarily designed for theory testing and model confirmation, PLS-SEM adopts a causal-predictive approach to SEM, focusing on maximising the variance explained in the model's dependent variables (Chin et al. 2020). Unlike CB-SEM, which requires stringent assumptions about data distribution and model complexity, PLS-SEM is robust to smaller sample sizes, non-normal data, and formatively specified constructs, making it ideal for estimating higher-order constructs and interaction effects, as is the case in our study (Ringle et al. 2020). While critics such as Rönkkö et al. (2016) argue that PLS-SEM may produce biased parameter estimates and lacks model fit indices compared to CB-SEM, these critiques overlook its predictive focus and its suitability for composite factor models in exploratory research (Henseler et al. 2014; Ringle et al. 2020). PLS-SEM's emphasis on prediction aligns with HRM research's applied goals (Ringle et al. 2020), which are often important for informing practical decisions, such as algorithmic HR choices.

Our research model further illustrates the necessity of PLS-SEM: it is a third-order hierarchical component model (HCM) comprising nine first-order reflective constructs, three second-order formative dimensions, and one third-order formative



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construct (i.e., HABMC), alongside two outcome constructs and a moderator—resulting in 16 latent constructs and 115 indicators, many of which are repeated across levels. CB-SEM presents significant challenges in accommodating such complexity, as it cannot directly model higher-order formative constructs. In contrast, PLS-SEM supports both formative and reflective specifications with greater flexibility and efficiency, while preserving the conceptual integrity of the model (Hair Jr et al. 2019). Moreover, the size of our model leads to a sharp increase in the number of parameters that must be estimated in CB-SEM, creating a risk of underidentification, convergence failures, or unstable solutions (Akter et al. 2017; Sarstedt et al. 2019). Importantly, our model employs the repeated indicator approach, whereby reflective indicators are reused at the first-, second-, and third-order levels—a method fully supported by PLS-SEM but not feasible in CB-SEM without violating core assumptions or requiring substantial structural compromises (Becker et al. 2012; Dijkstra 2010).

## 4 | Data Analysis and Results

Given the complexity of our model with higher-order constructs, we employed the repeated indicator approach using PLS-SEM to estimate the measurement and structural models and to confirm theoretical parsimony (Sarstedt et al. 2023). PLS-SEM has become a prevalent method in HR research for examining multiple constructs' direct, indirect, and moderating effects while accounting for measurement errors (Ringle et al. 2020). A critical review of 77 studies in HRM over 30 years concluded that PLS-SEM generates robust outcomes in objectively determining the impact of HRM practices on attitudinal, behavioural, and organisational outcomes (Ringle et al. 2020). Our study utilised SmartPLS 4.0 to estimate our proposed hierarchical model using nonparametric bootstrapping with 5000 samples to facilitate robust prediction, accurate factor identification, and reliable factor estimation for replications (Ringle et al. 2020). We employed both a priori and post-hoc methods to mitigate common method variance (CMV), in line with Hulland et al. (2018). Following the a priori method, we collected data from two distinct sample units to measure our study constructs. For the post-hoc method, we used theoretically unrelated variables as a marker variable. The correlation coefficient indicated a non-significant relationship between the variables ($r = -0.023$, $p > 0.05$).

### 4.1 | Measurement Model

The study assessed the measurement properties of all reflective first-order constructs (see Table 2). The dimensions HDABMC, HMOBMC, and HDPBMC form the third-order construct HABMC. The measurement model's results confirm the reliability of all scales, demonstrated by significant loadings of each item above the recommended threshold (0.70, $p < 0.001$), composite reliability (CR) scores above 0.80, and average variance extracted (AVE) surpassing the minimum threshold of 0.50 (Fornell and Larcker 1981).

To achieve discriminant validity, the square root of AVE of each latent construct should be greater than its correlation coefficient with any other latent construct. The square root of the AVEs shown on the diagonals of Table 3 confirms such validity (Fornell and Larcker 1981). We also analysed the cross-loadings of items across constructs, reaffirming that items of respective constructs had significantly higher loadings than those of other constructs. Further, discriminant validity was confirmed using Henseler et al. 's (2015) heterotrait-monotrait (HTMT) criterion (coefficients < 0.90). We examined the control variables using variance inflation factors (VIF). Our analysis reveals no collinearity, as all control variables indicated a VIF value much lower than the critical value of 5 (Ringle et al. 2020).

The results of our higher-order reflective measurement model, adhering to established guidelines (Sarstedt et al. 2019), are presented in Table 4. The path coefficients between first-order and second-order constructs are significant. Results reveal that COMP ($\beta = 0.92$), FORM ($\beta = 0.92$), and ACCU ($\beta = 0.91$) are significant subdimensions of HDABMC, with path coefficients significant at $p < 0.001$. Likewise, RELI ($\beta = 0.91$), FLEX ($\beta = 0.93$), and AMBI ($\beta = 0.90$) have significant relationships with the HMOBMC dimension, and COMPE ($\beta = 0.88$), BENE ($\beta = 0.90$), and INTE ($\beta = 0.93$) are significantly associated with HDPBMC dimension. These findings confirm the robustness of the second-order reflective model by demonstrating significant associations between second-order and first-order constructs.

### 4.2 | Structural Model

The results of our structural model (Table 5) report the significance of the hypothesised relationships through path coefficients ($\beta$) and the coefficient of determination ($R^2$). The findings confirm that HDABMC ($\beta = 0.40$, $p < 0.001$), HMOBMC ($\beta = 0.37$, $p < 0.001$), and HDPBMC ($\beta = 0.33$, $p < 0.001$) significantly and positively impact HABMC thus supporting H1, H2, and H3. Additionally, results show that HABMC significantly and positively influences AI-HPHRPs ($\beta = 0.75$, $p < 0.001$), supporting H4 while explaining 57% ($R^2$) of its variance. We also found that AI-HPHRPs have a significant positive impact on SHRE ($\beta = 0.57$, $p < 0.001$), thus supporting H5 while explaining 33% ($R^2$) of its variance. Furthermore, our findings show that TINT does not moderate the link between AI-HPHRP and SHRE ($\beta = 0.06$, $p = 0.515$), thus not supporting H6. Following the guidelines of Hayes et al. (2013) and using a bootstrapped sampling distribution with a 95% confidence interval, we confirm that AI-HPHRPs partially mediate the positive association between HABMC and SHRE. The direct effect of 0.75 reduces to 0.44, $p < 0.001$, in the presence of AI-HPHRPs (see Table 6). We examined the effect of the control variables based on similar previous studies in bias management (e.g., Akter et al. 2023) and HR-related firm-level outcomes (e.g., Gahan et al. 2021). The analysis showed that firm size ($\beta = -0.10$, $p = 0.074$), firm AI-integration ($\beta = -0.02$, $p = 0.706$), employee age ($\beta = -0.04$, $p = 0.386$), and employee gender ($\beta = -0.06$, $p = 0.215$) had no significant impact on SHRE.



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**TABLE 2** | Assessment of first-order, reflective model.

| Dimensions | Reflective constructs | Items | Loadings | CR | AVE |
|---|---|---|---|---|---|
| HR data bias management capability (HDABMC) | Completeness (COMP) | COMP1 | 0.90 | 0.90 | 0.74 |
| | | COMP2 | 0.85 | | |
| | | COMP3 | 0.83 | | |
| | Format (FORM) | FORM1 | 0.90 | 0.93 | 0.81 |
| | | FORM2 | 0.90 | | |
| | | FORM3 | 0.89 | | |
| | Accuracy (ACCU) | ACCU1 | 0.89 | 0.90 | 0.76 |
| | | ACCU2 | 0.84 | | |
| | | ACCU3 | 0.88 | | |
| Model bias management capability (HMOBMC) | Reliability (RELI) | RELI1 | 0.89 | 0.91 | 0.77 |
| | | RELI2 | 0.88 | | |
| | | RELI3 | 0.87 | | |
| | Flexibility (FLEX) | FLEX1 | 0.84 | 0.90 | 0.74 |
| | | FLEX2 | 0.87 | | |
| | | FLEX3 | 0.87 | | |
| | Ambidexterity (AMBI) | AMBI1 | 0.89 | 0.91 | 0.78 |
| | | AMBI2 | 0.88 | | |
| | | AMBI3 | 0.87 | | |
| Deployment bias management capability (HDPBMC) | Competence (COMPE) | COMPE1 | 0.77 | 0.89 | 0.73 |
| | | COMPE2 | 0.89 | | |
| | | COMPE3 | 0.90 | | |
| | Benevolence (BENE) | BENE1 | 0.79 | 0.87 | 0.70 |
| | | BENE2 | 0.85 | | |
| | | BENE3 | 0.86 | | |
| | Integrity (INTE) | INTE1 | 0.88 | 0.90 | 0.75 |
| | | INTE2 | 0.90 | | |
| | | INTE3 | 0.81 | | |
| Outcome constructs | AI-enabled HR practices (AI-HPHRP) | AI-HPHRP1 | 0.81 | 0.97 | 0.65 |
| | | AI-HPHRP2 | 0.81 | | |
| | | AI-HPHRP3 | 0.80 | | |
| | | AI-HPHRP4 | 0.72 | | |
| | | AI-HPHRP5 | 0.78 | | |
| | | AI-HPHRP6 | 0.79 | | |
| | | AI-HPHRP7 | 0.79 | | |
| | | AI-HPHRP8 | 0.81 | | |
| | | AI-HPHRP9 | 0.81 | | |
| | | AI-HPHRP10 | 0.83 | | |
| | | AI-HPHRP11 | 0.84 | | |
| | | AI-HPHRP12 | 0.82 | | |
| | | AI-HPHRP13 | 0.83 | | |
| | | AI-HPHRP14 | 0.84 | | |
| | | AI-HPHRP15 | 0.81 | | |
| | | AI-HPHRP16 | 0.82 | | |

(Continues)



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**TABLE 2** | (Continued)

| Dimensions | Reflective constructs | Items | Loadings | CR | AVE |
|---|---|---|---|---|---|
| | | AI-HPHRP17 | 0.77 | | |
| | | AI-HPHRP18 | 0.80 | | |
| | | AI-HPHRP19 | 0.83 | | |
| | | AI-HPHRP20 | 0.81 | | |
| | | AI-HPHRP21 | 0.82 | | |
| | Strategic HR effective (SHRE) | SHRE1 | 0.86 | 0.95 | 0.73 |
| | | SHRE2 | 0.88 | | |
| | | SHRE3 | 0.86 | | |
| | | SHRE4 | 0.82 | | |
| | | SHRE5 | 0.86 | | |
| | | SHRE6 | 0.86 | | |
| | | SHRE7 | 0.84 | | |
| | | SHRE8 | 0.85 | | |

## 5 | Discussion

### 5.1 | Summary of Findings

Our study identifies three critical dimensions of algorithmic bias management capability in HR: HR data bias, HR model bias, and HR deployment bias management capability. Each dimension comprises three subdimensions that collectively influence the overall strategic effectiveness of AI-driven HR systems.

Firstly, we found that HR data bias management capability, which focuses on data completeness, format, and accuracy, significantly impacts overall HABMC, confirming H1. This highlights the importance of using comprehensive and well-formatted training data to develop unbiased HR algorithms. For instance, Google's success with AI-enabled models is attributed to the high quality of its training data, leading to fair and effective outcomes (Google 2024). Conversely, Amazon's recruitment algorithms showed bias against women due to flawed training data, reinforcing the need for accurate data in HR decisions (Dastin 2018; Rodgers et al. 2023).

Secondly, HR model bias management capability, characterised by model reliability, flexibility, and ambidexterity, also significantly influences HABMC, supporting H2. Reliable models ensure unbiased candidate selection, while flexibility allows HR practices to adapt to changing business environments. Ambidexterity enables HR professionals to optimise existing processes and explore new methods simultaneously.

Thirdly, HR deployment bias management capability, which includes the competence, benevolence, and integrity of AHR systems, plays a crucial role in preventing and correcting biases during deployment, confirming H3. Continuous monitoring and adjustment of algorithms ensure fair outcomes in real-world HR applications.

Our results further suggest that HABMC positively impacts AI-enabled high-performance HR practices, which, in turn, enhance strategic HR effectiveness, confirming H4 and H5. This alignment fosters trust in ethical algorithmic decisions, boosting employee morale and retention (M. K. Lee 2018). However, the moderating effect of employee turnover intention (H6) was not supported, indicating that in robust HR ecosystems, individual turnover intentions have less impact on strategic outcomes (Guest and Conway 2011; Huselid et al. 1997). This finding indicates that organisational outcomes are mainly shaped by systemic factors, such as AI-enabled HPHRPs, rather than by individual-level perceptions alone. Additionally, the context of the finance and insurance industry, where AI-enabled HR practices tend to be standardised, may explain why turnover intentions play a limited role in influencing strategic HR outcomes in such structured environments.

### 5.2 | Theoretical Implications

With the increasing application of AI in the workplace, researchers underscore the importance of developing distinct DCs to effectively leverage algorithms for decision-making and succeed in dynamic business environments (Gupta et al. 2020; Simón et al. 2024). Moreover, there remains a dearth of research on DCs in HRM and their impact on firm resources, practices, and outcomes, alongside calls to integrate the DC perspective to advance SHRM scholarship (Apascaritei and Elvira 2022; Chadwick and Flinchbaugh 2021; Chatterji and Patro 2014). Responding to these calls and drawing on the DC lens, our study proposes a comprehensive bias management capability framework that identifies HR data, model, and deployment bias capabilities as core components of dynamic HRM capabilities. Building upon and extending the work of Akter et al. (2023), which examined AI bias in industrial marketing, we apply these insights to the HR domain to explore their effects on firm-level HR outcomes. While interest in examining AI bias within HR scholarship is growing (Charlwood and Guenole 2022), our study is pioneering in developing and applying these capabilities in this context. These capabilities enable managers to sense and seize market opportunities, mitigate threats, and dynamically



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**TABLE 3** | Correlations of LVs, AVEs and descriptive statistics[a].

| Constructs | Mean | SD | COMP | FORM | ACCU | RELI | FLEX | AMBI | COMPE | BENE | INTE | AI-HPHRP | TINT | SHRE | Age | Gender | Firm size | AIINT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Completeness (COMP) | 4.86 | 1.13 | **0.86** | | | | | | | | | | | | | | | |
| Format (FORM) | 4.26 | 0.73 | 0.77 | **0.90** | | | | | | | | | | | | | | |
| Accuracy (ACCU) | 5.09 | 1.04 | 0.76 | 0.73 | **0.87** | | | | | | | | | | | | | |
| Reliability (RELI) | 4.94 | 1.06 | 0.74 | 0.68 | 0.76 | **0.87** | | | | | | | | | | | | |
| Flexibility (FLEX) | 5.11 | 1.03 | 0.63 | 0.61 | 0.69 | 0.78 | **0.86** | | | | | | | | | | | |
| Ambidexterity (AMBI) | 5.04 | 1.07 | 0.60 | 0.57 | 0.64 | 0.69 | 0.75 | **0.88** | | | | | | | | | | |
| Competence (COMPE) | 5.17 | 1.15 | 0.66 | 0.70 | 0.65 | 0.63 | 0.60 | 0.57 | **0.85** | | | | | | | | | |
| Benevolence (BENE) | 5.25 | 1.02 | 0.57 | 0.53 | 0.62 | 0.60 | 0.57 | 0.56 | 0.67 | **0.83** | | | | | | | | |
| Integrity (INTE) | 5.25 | 1.13 | 0.58 | 0.58 | 0.61 | 0.61 | 0.54 | 0.54 | 0.72 | 0.79 | **0.86** | | | | | | | |
| AI-enabled HR practices (AI-HPHRP) | 4.8 | 1.2 | 0.73 | 0.62 | 0.64 | 0.59 | 0.63 | 0.62 | 0.61 | 0.55 | 0.57 | **0.80** | | | | | | |
| Turnover intention (TINT) | 3.91 | 1.61 | −0.06 | −0.07 | −0.12 | −0.06 | −0.03 | −0.01 | −0.03 | −0.12 | −0.10 | −0.09 | **0.80** | | | | | |
| Strategic HR effectiveness (SHRE) | 5.39 | 1.15 | 0.55 | 0.57 | 0.62 | 0.59 | 0.50 | 0.48 | 0.71 | 0.74 | 0.73 | 0.57 | −0.08 | **0.85** | | | | |
| Age | NA | NA | 0.00 | 0.00 | 0.03 | −0.00 | −0.01 | −0.07 | −0.02 | −0.01 | −0.01 | −0.04 | −0.07 | −0.05 | **1** | | | |
| Gender | NA | NA | 0.02 | 0.04 | −0.00 | 0.02 | −0.01 | 0.03 | −0.02 | −0.04 | −0.06 | 0.02 | 0.07 | −0.04 | −0.17 | **1** | | |
| Firm size | NA | NA | 0.04 | −0.01 | −0.02 | 0.01 | −0.03 | 0.05 | −0.02 | −0.01 | −0.03 | −0.07 | −0.07 | −0.13 | −0.04 | 0.01 | **1** | |
| AI integration (AIINT) | NA | NA | 0.37 | 0.28 | 0.31 | 0.22 | 0.26 | 0.29 | 0.32 | 0.24 | 0.21 | 0.39 | 0.01 | 0.18 | 0.02 | −0.04 | 0.20 | **1** |

[a]Square root of AVE (in bold) is reported on the diagonal. Values of 1 for control variables reflect the square root of AVE for single-item constructs. Mean and standard deviation are not reported for control variables, as they are measured on nominal scales.



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**TABLE 4** | Assessment of the higher-order model.

| Model | Second-order | First-order | $R^2$ | $\beta$ | t-statistics |
|---|---|---|---|---|---|
| HR algorithmic bias management capability | HDABMC | COMP | 0.85 | 0.92 | 88.93 |
| | | FORM | 0.84 | 0.92 | 66.33 |
| | | ACCU | 0.82 | 0.91 | 70.66 |
| | HMOBMC | RELI | 0.82 | 0.91 | 64.79 |
| | | FLEX | 0.86 | 0.93 | 75.65 |
| | | AMBI | 0.80 | 0.90 | 45.81 |
| | HDPBMC | COMPE | 0.78 | 0.88 | 43.28 |
| | | BENE | 0.82 | 0.90 | 48.43 |
| | | INTE | 0.87 | 0.93 | 78.10 |

Abbreviations: ACCU, accuracy; AMBI, ambidexterity; BENE, benevolence; COMP, completeness; COMPE, competence; FORM, forma; FLEX, flexibility; HDABMC, HR data bias management capability; HMOBMC, model bias management capability; HDPBMC, deployment bias management capability; INTE, integrity; RELI, reliability.

**TABLE 5** | Results of the structural model.

| Hypotheses | Main model | Path coefficients | Stand error | t-stat |
|---|---|---|---|---|
| H1 | HDABMC > HABMC | 0.40 | 0.016 | 25.25 |
| H2 | HMOBMC > HABMC | 0.37 | 0.014 | 26.14 |
| H3 | HDPBMC > HABMC | 0.33 | 0.017 | 19.40 |
| H4 | HABMC > AI-HPHRP | 0.75 | 0.034 | 22.21 |
| H5 | AI-HPHRP > SHRE | 0.57 | 0.057 | 10.02 |
| H6 | TINTxAI-HPHRP > SHRE | 0.07 | 0.098 | 0.73 |

Abbreviations: AI-HPHRP, AI-enabled HR practices; HABMC, HR algorithmic bias management capability; HDABMC, HR data bias management capability; HDPBMC, deployment bias management capability; HMOBMC. model bias management capability; SHRE, strategic HR effectiveness, TINT, turnover intention.

**TABLE 6** | Results of the mediation testing.

| Hypothesis | Mediating path | Indirect effect | T-value | Significance | CI 5% | CI 95% |
|---|---|---|---|---|---|---|
| H7 | HABMC > AI-HPHRP > SHRE | 0.44 | 7.019 | ($p < 0.001$) | 0.34 | 0.54 |

Abbreviations: AI-HPHRP, AI-enabled HR practices; HABMC, HR algorithmic bias management capability; SHRE, strategic HR effectiveness.

reconfigure the firm's human resource base by addressing biases in AI-driven HR systems.

We contribute to the ongoing debate regarding the conditions and pathways that enable HRM to assume a strategic role within organisations (Jiang and Messersmith 2018). By integrating insights from DC theory and SHRM literature, we illuminate the DCs associated with AHR bias management, offering theoretical explanations and empirical evidence to establish the impact of HABMC on AI-HPHRPs and strategic HRM effectiveness. Our work extends existing research highlighting the potential of firm resources and capabilities, along with effective HR systems, to drive superior organisational outcomes (Boxall and Macky 2009; Delery and Roumpi 2017). Our study broadens the strategic scope of HRM by shifting the focus from traditional employee behavioural impacts to firm-level capabilities and outcomes—a dimension largely neglected in prior research (Apascaritei and Elvira 2022; Chadwick and Flinchbaugh 2021; Jiang et al. 2013), while exploring the broader implications of DCs beyond firm performance. The proposed framework also demonstrates how organisations can derive strategic value from HR systems despite the ethical challenges of AI, such as biases. This also contributes to the discourse on managing the dark side of AI-driven HR practices (Zhou et al. 2023; Vassilopoulou et al. 2024).

A persistent debate in the HPHRPs literature centres on whether researchers adequately address the context, boundary conditions, and complementarities under which these HR systems and practices are expected to be effective (Gahan et al. 2021). We focus on AI-enabled HPHRPs, a novel phenomenon emerging within the context of algorithmic people management. Our empirical findings establish that AI-enabled HPHRPs must be supplemented by HABMCs to be effective, as algorithm-based practices can be perceived as unfair and biased. Building on prior studies that examine the impact of HPHRPs on firm innovation and performance (e.g., Haar et al. 2021), our research contributes to this literature by highlighting the influence on strategic HR effectiveness. This critical organisational outcome has been underexplored. Additionally, we expand the literature by exploring the mediating role of AI-enabled HPHRPs and examining other intervening factors, such as employee perceptions of leaving the firm, that may influence their strategic outcomes.



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### 5.3 | Practical and Policy Implications

Developing new DCs to navigate the shifting business landscape and address conflicting demands has become imperative for firms (Chadwick and Flinchbaugh 2021; Schilke et al. 2018). HABMC enables organisations to adjust practices rapidly, adapt to evolving environmental conditions and disruptions, and drive change through an iterative and developmental approach to AI-powered people management. Building on the HABMC framework, we emphasise the critical importance of enhanced data management capabilities, ensuring that datasets are comprehensive, accurate, and well-structured. This should be complemented by AI model development capabilities to improve the reliability, flexibility, and ambidexterity of AI systems, ensuring HR algorithms are robust, adaptable to diverse populations, and responsive to changing organisational needs. Our model outlines a roadmap for organisational leaders to implement effective AI and data governance policies for workforce management. This includes regular audits, updates, and validation processes to eliminate biases. For example, IBM's bias audits involve a diverse evaluation team that reviews the entire hiring process, from sourcing to final selection, ensuring that all steps are free from biases related to social status, religion, sexual orientation, subcultures, age groups, gender, and other social categories (IBM 2023).

With the adoption of AI, the need to design HRM systems coherently has become more important than ever. HRM systems that build, integrate, and reconfigure the human capital base must be implemented with careful consideration of existing practices, their interconnected effects, and their overall alignment with other organisational systems and resources, such as technology (Boxall and Macky 2009; Gahan et al. 2021). This highlights the importance of HR managers, who are constantly challenged to improve the effectiveness of technology-driven HR systems, allocating organisational resources toward developing capabilities to manage undesirable outcomes, such as biases, to ensure impartial HR outcomes. According to our deployment bias management capability, firms should develop the necessary skills and knowledge base to responsibly deploy AI models in HR functions while upholding ethical standards to build trust, especially among employees who may be vulnerable to algorithmic HR decisions (Rodgers et al. 2023). For example, organisations can provide targeted training for HR leaders and data practitioners to develop bias detection and management competencies. By doing so, firms can ensure that HR systems contribute to the strategic needs of HR and the organisation, as indicated by the results of this study. Thus, developing HABMC is a strategic necessity.

At a broader level, our research addresses the growing societal demand for inclusive, diverse workplaces and humanistic, ethical approaches to people management (Rodgers et al. 2023; Zhou et al. 2023). The integration of AI into workforce management has created significant tensions. While AI can provide unprecedented actionable insights, ethical concerns often raise questions about the fairness and acceptability of these outcomes. By applying our framework, firms can simultaneously mitigate AI biases, optimise AI-driven HR practices, and achieve strategic HR objectives. In line with prior research, a well-designed HRM system not only enhances employee well-being but also supports the integration of work-life balance practices and fosters positive employment relationships by prioritising fairness and building trust (Guest 2017).

### 5.4 | Future Research and Conclusions

This study has limitations, which also provide avenues for future research. First, while we identified key dimensions and subdimensions of HABMC, it is important to highlight that strategic HR practices and organisational changes are shaped by a constantly changing dynamic environment, particularly in the wake of new technologies. The relevance and impact of these subdimensions may shift over time. Thus, longitudinal studies should examine how these dimensions and subdimensions evolve and assess their long-term effectiveness in managing algorithmic biases for strategic people management.

Second, while our model was empirically tested and found effective in the Australian insurance and finance industries, its generalisability could be enhanced by conducting further empirical studies across different industries, organisational and country contexts. This would help validate our findings and determine the applicability of our model in diverse settings, ensuring broader relevance and utility beyond the HR domain.

Additionally, given the interconnected nature of today's workplaces, it would be valuable to investigate how HABMCs influence employee outcomes such as employee performance and well-being and other broader organisational outcomes.

In conclusion, our study highlights the critical role of HABMC in shaping AI-enabled HPHRPs and their impact on strategic HRM effectiveness. Organisations can foster fairer and more effective HR practices by addressing data, model, and deployment biases. Future research should continue to explore and refine these capabilities, ensuring that AI technologies contribute positively to organisational strategic goals and societal well-being.

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The authors have nothing to report.

#### Conflicts of Interest

The authors declare no conflicts of interest.

#### Data Availability Statement

The data that has been used is confidential.

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